



Proposal for Research

Title:

**An analysis of algae-based bioplastics for use in
self-replicating machines.**

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Project Summary

This research project explores the use of algae-based bioplastics for its application as a core material in self replicating machines. Which are devices that reproduce using common materials. Algae, chosen for its rapid growth and adaptability, will be used to produce glycerol and starch from promising strains (*Chlamydomonas reinhardtii*, *Chlorella sorokiniana*, and *Botryococcus braunii*). These compounds will form bioplastics. Then tested for strength and 3D printability to assess their potential in replicating machine parts. Success could support sustainable, decentralized manufacturing with applications ranging from aiding in rural development to space exploration.

Background Information

Ever since the machine was invented by man, it has been wondered if a machine could reproduce as man does. This is the idea of self-replicating machines, whereby using raw materials around it a machine can both produce useful goods alongside copies of itself. Yet even in the modern age such a system has only partially arrived. The aim of this study is to test current and theorized methods to aid in achieving self replication.

As of the writing of this paper, no fully replicating system has been deployed. Though a wide variety have been theorized in research papers. Many aim to solve everything from CO2 emissions to large scale space colonization (Ellery, 2016). While Most papers point out the exponential potential of replicating machines, many fail to note that there exists a similar recursive issue when designing the system. Every complexity added to the process of creation increases the time in which it takes for each replication and failure rate of replication. Therefore lots of untraditional systems will have to be conceptualized and tested to maximize the simplicity and probability of successful replication. Unfortunately most of these papers are written by young students and therefore have a high level overview of the topic. While they propose solutions there is very little testing or other feasible proof of their ideas or methods. One organization that is doing real hardware testing for replication is RepRap, an open source 3d printer. The

RepRap project could be seen as the highest level of replication achieved by a modern system, with some limitations being that it can print only plastic parts of itself acquired externally by a human user.

This is far from the desired state of a fully self replicating machine which would not only require plastic that would ideally be created internally and automatically, but also extremely complex systems of motors, sensors and circuits boards, computation, alongside memory storage and transfer. Though for the scope of this project I will focus on the creation of plastic by the system itself. Specifically on the creation of plastic from scratch so that it can be used as a base material for replicating machines, This requires the process to be relatively simple, efficient, requiring little human intervention alongside having to create plastic with as many environmental factors as possible. It is for these reasons that bioplastic has been chosen. Bioplastics are a polymer based formulation that is commonly derived from renewable biomass sources, such as plants, algae, or microorganisms (Mathiot, 2018). This paper will explore using algae as a source of biomass due to algae's properties of rapid growth, high photosynthetic efficiency and environmental adaptability (Mathiot, 2018). Specifically 3 strains, *Chlamydomonas reinhardtii* and *Chlorella sorokiniana* for their rapid growth and efficiency in creating Starch (Mathiot, 2018) and *Botryococcus braunii* for its high content in glycerol-rich lipids (Chisti, 2007). These strains will eventually be pressed into an oil and saponified from a sodium hydroxide sample based on common salt in order to create glycerol. The glycerol and starch of each sample will then be divided equally and combined with itself and others to create a variety of Bioplastics, which will finally be tested both as a general plastic and as a 3d printable material.

Significance of research

If the presented methods and testing do ultimately prove feasible it could aid in the creation of a fully replicating system. When such a system arises even its most basic form would allow for an extremely decentralized approach to manufacturing. Allowing people to buy raw materials or find them locally and produce a good therefore increasing supply and reducing the cost of goods alongside increasing the speed and industrialization of foreign nations. Imagine enabling the production of goods

ranging from a latch for an old gates fence to an electric fan to provide cooling, all grown from algae or common materials in poor rural communities.

Question to be Studied

The focus of this research is to investigate the potential use of bioplastics as a core material for self-replicating machines, specifically based on algae. Algae is known for its efficiency among plants for photosynthesis and its rapid growth, Making it a great resource for bioplastic production (Mathiot, 2018). The proposed hypothesis is that bioplastics created from some strains of high starch algae possess the mechanical properties required for use in the fabrication of self replicating machines, Allowing for significantly reduced costs and more sustainability in many areas of everyday life

Research Plan

Objective I: To determine whether algae oil can be utilized in bioplastic production

Rationale: This analysis will include the application of heat and mechanical pressure onto many species of algae, then lab-grade NaOH will be added to form glycerol. Furthermore traditional starch, vinegar, and water will successfully create a bioplastic. Proving that algae based oil is feasible for bioplastic production.

Objective II: To determine whether algae based oil and it's own starch is viable for bioplastic production

Rationale: This analysis will include the application of heat and mechanical pressure on to many species of algae, the storage of the remaining algae and the disruption of algae cell walls to release starch through various methods (chem, freeze, mechanical). Finally the addition of lab grade NaOH to form glycerol further added to vinegar and water and finally the extracted starch successfully creates a bioplastic

Objective III: To determine whether algae based oil and it's own starch in combination with NaOH electrolysis is viable for bioplastic production

Rationale: This analysis will focus on the Production of NaOH, which is required for bioplastic production, synthesized from the much more common NaCl. This is possible due to electrolysis of NaCl this allows for the formation of NaOH, which can then filter out from the solution.

Objective IV: To determine if algae based bioplastics are statically comparable in strength to other plastics.

Rationale: Ensuring that the algae based plastic is similar in strength to other common plastics is essential for its use in self-replicating machines. If it's too weak, it will not be able to support the weight of other components or stand up to external forces. This analysis will use tensile testing machines alongside physics calculations for both algae based plastics and other common plastics to see how they compare in strength.

Objective V: Algae-based bioplastic is able to be extruded for use in self replication and retains strength comparable to to other plastics

Rationale: Ensuring that algae-based bioplastic can be utilized in a 3D printing system is fundamental to self-replicating machines. This analysis will involve creating an extruder, the extrusion process, and testing the tensile strength of resulting 3D prints using commercially available 3D printers.

Experimental Design

Saponification of fats: Is the process which will be used to convert the loose triglycerides of algae oil in combination with sodium hydroxide into a glycerol alongside a resulting waste product of a crude soap. While the exact amounts will vary depending on the objective, the ratio is roughly 8 grams of sodium hydroxide for every 60 grams of the algae oil, which will yield roughly 5mL of glycerin at that ratio.

Algae Growth: the algae will be grown within 6, 1 gallon plastic tanks which will be 3d printed using polylactic acid based plastic and have an interior volume of 4,096,000 mm cubed. Each tank will contain a sample of algae, as there are 3 samples every sample will have two tanks dedicated to it, and algae is never to be cross contaminated. Every day the tanks will be irrigated and stored to ensure that the most amount of algae experiences growth and therefore a higher yield occurs. All species will be grown

continually over a period of roughly 6 months with the required amount of algae for a given objective removed and processed right before it is required.

Oil Processing: A manual Oil Press will turn the grown algae into algae oil and algae “starch” through a process of mechanical force and heat. Based on the growth rate of a typical chlorella strain an estimated 1.969mL per day of oil can be grown each day under average conditions. Leading to a total sample of roughly 400mL of oil across the 3 species.

Starch processing: the remaining starch from each algae sample will be stored by species and then cooled and then finally mechanically pressed again before its use in testing after objective 1. This cooling and mechanical force is to ensure that the cell walls have been broken leading to the release of the internal starch.

Sodium hydroxide (NaOH) electrolysis: Is the process of turning a basic salt, NaCl using electrical power applied through two electrodes in distilled water into NaOH which is formed on the negative electrode. This will be done using makeshift graphite electrodes from common pencil lead on either side of a 3d printed container that will allow for the slow creation of NaOH over several hours. The ideal yield is roughly 31 grams for every 50 grams of NaCl based on the chemical formula if all the NaCl is processed. This electrolysis reaction will be performed under a fume hood as a small amount of hydrogen and oxygen gas will be produced due to the splitting of water.

3D printing: Is a method of manufacturing which allows for the excursion of material typically plastics based on motor control traditionally along 3 axes and sensors to detect its current position in space. A commonly available commercial 3d printer will be used for the algae based plastic and

Tensile Strength Testing: This will be done using weights with a known mass in combination with gravity in order to apply an easily calculable force upon several plastic parts of constant shape and volume, the weights will be applied progressively and until the point of mechanical failure. The measurement itself will be done through digital force measurement devices and supported by physics calculations to ensure that the devices are recording accurate data.

Plastic Extruder: This is a device that will have to be crafted to fulfill objective 5. Plastic extruders commonly work by taking a large sample of plastic of un-uniform shape and based on the grinder mechanism they are melted and grinded into a certain size and general shape referred to as pellets. These pellets then fall between the groves of a rotating screw and are then heated and rotated to the end tip with a specific diameter of roughly 1.75mm which is what is accepted by commercial 3D printers. It exits as a line of plastic which is then rapidly cooled repeatedly until the initial sample is fully processed. Once it has fully cooled it is referred to as filament and is able to be used in various commercially available 3D printers, enabling the results and insights found from objective 5.

Risk and Safety

The main risks and hazards involved in this project are Sodium hydroxide's properties as a strong base (NaOH), Electrical shortage of the extruder or Injury through the extruder's heating or crushing elements.

To mitigate the risks associated with the materials and equipment involved, appropriate personal protective equipment (PPE), including a laboratory coat, eye protection, and nitrile gloves, will be worn at all times in the laboratory. Additionally, an N-95 mask will be worn when handling sodium hydroxide in powdered form to prevent inhalation. When handling heavy components of the extruder, steel-toed boots will be required to prevent foot injuries. Nichrome will not come into contact with anyone's skins for a prolonged period due to possible allergic skin reactions in those who are sensitive to nickel (ESPI metals, 2014). Sodium hydroxide (NaOH) will be kept separated from any sources of water when not in use to avoid accidental exothermic reactions (renae, 2022). Before transitioning the system from a building state to a testing state, it will be fully disconnected from all electrical power sources to prevent electrical hazards.

For disposal procedures, biodegradable materials such as algae, glycerol, oil, or the final bioplastic (if produced in excess or deemed a failure) will be safely disposed of through traditional disposal methods. Chemical waste, however, will follow the lab's chemical hygiene plan, which requires labeling

all waste containers by contents and concentrations when possible. The final handling and disposal of hazardous chemical waste will be managed by the lab's designated outside vendor.

ESPI metals. (2014). *Nickel-Chromium Alloy* | | *ESPI Metals Specialists - ESPI Metals*.

[Www.espimetals.com](http://www.espimetals.com).

<https://www.espimetals.com/index.php/msds/974-Nickel%20Chromium%20Alloy>

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